
When Are Recursive Language Models Useful? A Cost-Aware, Task-Conditional Evaluation

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Abstract

1 Recursive Language Models (RLMs) handle long inputs through controller-
2 generated programs and additional model calls. Published RLM evaluations
3 typically compare these scaffolds with generic long-context or coding baselines.
4 Deployment requires a different comparison: whether adaptivity improves on
5 routes suited to the task under a common information, scoring and budget contract.
6 We evaluate the released standard RLM against deterministic operators, retrieval,
7 semantic workers and direct inference in an operation-matched audit. On controlled
8 operations and known-family Oolong aggregation, matched routes generally
9 match or outperform standard RLM at lower cost. A validation-source replication
10 preserves this direction although its clustered exact-rate difference includes zero.
11 Open multi-document search is less stable. RLM beats BM25 on a first untouched
12 45-query shard, but neither a prospectively locked equal-cap comparison nor a
13 same-query repeat yields an RLM contrast that survives multiplicity correction.
14 Route ordering changes across these rollouts while RLM remains less reliable
15 and 5.8–15.9 $\times$ costlier than the stronger alternatives. Component controls distinguish
16 operation choice, answer formatting and execution failures. A comparison
17 restricted to completed queries can reverse the observed ordering, so route choice
18 must be based on all attempted queries. These results support evaluating recursive
19 inference through task-specific accuracy, cost and completion, with any positive
20 deployment advantage established against a strong matched alternative.

21 1 Introduction

22 Long-context systems can retrieve, compress, apply fixed operators, delegate local judgments,
23 use direct inference or generate adaptive programs. Strong benchmark performance by adaptive
24 programming establishes capability but not deployment value. A deployment decision also depends
25 on the information and tools available to each route, cost, completion and ranking stability.

26 RLMs [Zhang et al., 2026] make this distinction consequential. A controller writes Python over
27 externalized context and may recursively query models. The current RLM paper compares task-
28 agnostic long-context and coding scaffolds, varies recursion depth and studies post-training; it reports
29 strong OOLONG-Pairs results [Zhang et al., 2026, Bertsch et al., 2026]; follow-ups add self-reflective
30 search, typed combinators and recursive harnesses [Alizadeh et al., 2026, Roy et al., 2026, Lumer
31 et al., 2026]. Deployed alternatives include RAG–long-context routing [Li et al., 2024, Chen et al.,
32 2026a], retrieval [Lewis et al., 2020], prompt compression [Jiang et al., 2023], deterministic operators
33 and decomposed workers [Khot et al., 2023]. The relevant comparison is RLM against the strongest
34 affordable route permitted by the task contract.

35 We evaluate this deployment decision directly. For a task family (a shared operation and answer
36 schema), an information contract (the context, tools and prior knowledge available to each route) and

a budget (calls or decisions, time and measured cost), we ask whether adaptive program generation yields a stable accuracy–cost advantage over operation-matched alternatives. This extends cost-aware agent auditing [Kapoor et al., 2025, 2026, Bragg et al., 2026] and differs from RLM trajectory diagnosis: the unit is the deployed route, not recursion depth or one scaffold component. Declared specialization, paired uncertainty and rollout stability determine the credibility of the resulting task-conditional route frontier [Yu and Kumbier, 2020].

We compare released standard RLM with deterministic, retrieval, semantic-worker and direct routes. Oolong and an RLM trajectory already decompose labeling from aggregation [Zhang et al., 2026, App. E.2]; our study adds a same-row comparison of complete deployed routes. Table 3 summarizes the boundary. We contribute:

1. An operation-matched protocol that fixes rows and scorers, declares route-specific prior knowledge and reports accuracy, failures, latency, calls and cost. An answer-contract repair separates finalization failure from route quality.
2. Wrong-family, same-operator, finalization and trace controls that distinguish operation knowledge from tool access and localize differences in semantic judgment, output schema and scaffold reliability; prior RLM trajectory diagnoses do not combine these controls [Zhang et al., 2026, Wang, 2026].
3. A bounded deployment comparison: known-operation routes lead on the evaluated Oolong slices, while adaptive-search differences remain unresolved. Paired rollout and completion analyses show how the same outputs can support different apparent rankings when the evaluation population changes.

2 Operation-Matched Evaluation

Scope. The audited system is the official `r1ms v0.1.3` inference-time scaffold: a controller generates Python over externalized context and may call additional models before answering. We compare this implementation with task-matched routes on identical instances. T-STRUCT, T-MIXED and Oolong have direct standard-RLM comparisons; T-NLP is diagnostic. Tested controller changes and official trained-policy/code checks probe robustness, but do not reproduce the original SRLM, λ -RLM or RAH benchmark protocols. Conclusions remain specific to the evaluated implementation and settings.

Tasks and independence. Table 1 defines three controlled families: exact structured aggregation (T-STRUCT), directional semantic counting (T-MIXED) and standard classification (T-NLP, diagnostic only). Oolong separates local semantic labeling from aggregation. Label-provided slices (LP-80 and LP-150) isolate aggregation and answer formatting rather than end-to-end performance. The standard-unlabeled slice (SU-150) removes those labels but reuses development contexts, so it supplies scale evidence. Two 45-case slices test different limits: the context-disjoint slice (CD-45) excludes prior context windows and spans 1k, 4k and 262k tokens; the validation-source slice (VC-45) uses two previously unused corpora from the official validation split. Oolong supplies no source-document IDs, so neither slice establishes document-level independence.

BrowseComp-Plus tests search in fixed pools of 1,000 documents per query [Chen et al., 2026b]. The original 80-query study supplies exploratory evidence; a first untouched 45-query shard tests replication. A second untouched 45-query shard adds iterative search and common decision/time caps before scoring. A later seed repeat reuses those same queries and is a rollout sensitivity, not 45 new independent cases. LongBench-v2 Academic QA ($N=94$) remains a separate format-validity diagnostic [Bai et al., 2025].

What the typed route does. Consider an Oolong question asking whether one label is more frequent than another for a named user. The worker labels individual records; fixed code filters that user’s records, counts the two labels and formats their relation. The route knows the operator family but receives no gold local labels. An RLM must instead choose and execute its program. Comparing complete routes therefore tests the value of that additional adaptivity under declared specialization.

Evaluation design. Figure 1 specifies the information available to each route. A complete route includes input handling, model or tool calls, answer finalization and failures. Task-specific operators

Table 1: Controlled task-family construction. Each family targets a different long-context operation with a prespecified alternative route.

Family	Operation	N	Expected cheap mechanism
T-STRUCT	Exact pair enumeration	30	SLM extraction + fixed operator
T-MIXED	Directional semantic counting	30	Direct LLM / local judgments
T-NLP (diagnostic)	Standard classification	24	SLM direct

Table 2: Model roles and their corresponding evaluation settings.

Role	Model	Use
RLM ctrl.	Gemini 2.5 Flash; GPT-5; Qwen3.6-35B-A3B	Standard scaffold; direct matched Gemini; BrowseComp matched routes
Trained/code RLM	Qwen3-30B-A3B; λ/Qwen3-8B	Frozen CD-15 diagnostics
Direct LLM	Gemini 2.5 Flash; Qwen3.6-35B-A3B	Direct-context and matched-controller baselines
SLM worker	Gemma 4 26B-A4B	Direct, CoT, labeling and fallback routes
Cheap controller	GPT-4o-mini	Deliberately weak cheap controller; T-NLP orchestration only

and answer schemas are intentional deployment specialization, so comparisons are not pure scaffold effects. Wrong-family, same-tools and finalizer controls test whether an observed advantage comes from operation knowledge, access to tools or answer formatting. Frozen rows, scorer versions, decision/time caps and failure records link each result to its comparison contract.

Baselines and model roles. Table 2 identifies the models. The primary Oolong RLM controller is Gemini 2.5 Flash; T-STRUCT uses GPT-5. All BrowseComp routes use Qwen3.6-35B-A3B. There, the text route issues one batch of textual subqueries, while iterative search updates its lexical query after each retrieved batch; neither generates and executes Python. Within each endpoint, decoding is matched: the fixed-45 and equal-cap endpoints use temperature 0.7, seed 20 and thinking disabled; the repeat changes only the seed to 21. The original 80-query endpoint and both judges use temperature 0. Self-hosted models are pinned by revision and container digest. Appendix H gives the full settings, hosted-model identifiers and protocol differences.

Evidence roles. LP isolates aggregation; SU tests end-to-end scale on reused contexts; VC tests unused-source transfer; and CD tests specialization and shared tools. For search, the prospectively locked equal-cap endpoint is primary, the two earlier studies provide context and the same-query seed repeat is a rollout sensitivity. These roles are not pooled.

Estimands, scoring and uncertainty. For paired routes r, s on n rows, $\hat{\Delta}_A = n^{-1} \sum_i (Y_{ir} - Y_{is})$ and $\hat{\Delta}_C = n^{-1} \sum_i (C_{ir} - C_{is})$; failed or invalid executions have $Y = 0$. A route is empirically dominated if another has no lower observed accuracy and lower cost. This point-estimate relation does not resolve uncertainty or supply a deployment-specific value for accuracy.

Paired accuracy comparisons use exact McNemar tests; b, c denote first-only and second-only correct. BrowseComp difference intervals use 50,000 paired-row bootstrap draws. Oolong intervals resample context-window clusters (5,000 draws): 72 for development $N=120$, 73 for SU-150, 32 for CD-

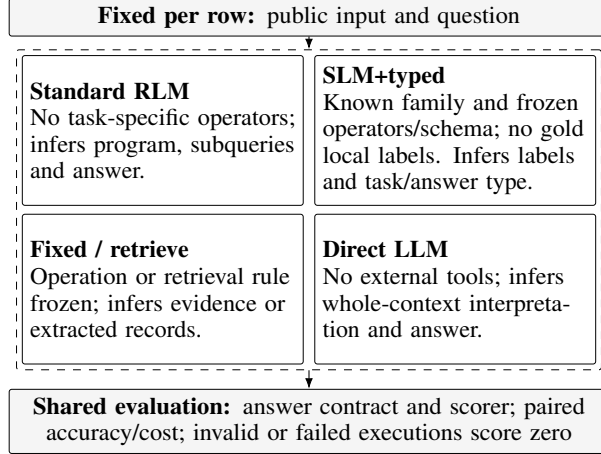


Figure 1: Route and information contracts. Allowed deployment prior knowledge is declared before comparison. The estimates compare complete deployed routes. They do not isolate a scaffold-only causal effect or evaluate an automatic router.

45 and 20 for VC-45. Intervals are conditional on observed executions. Within each equal-cap rollout, Holm correction covers the three RLM comparisons against BM25, text and iterative search. Subgroup, controller and mechanism probes are descriptive; there is no global familywise claim.

Oolong reports binary exact accuracy and the benchmark’s partial-credit mean score: numeric near-misses receive $0.75^{|\Delta|}$ and relative-frequency answers accept the prescribed relation substrings. BrowseComp uses a method-blinded Gemma 4 semantic judge and a separate GPT-5.6 audit. Both compare predictions with benchmark gold answers; cross-judge agreement measures stability, not human validity. MCQ diagnostics map predictions back after answer shuffling.

Cost accounting. RLM cost is decomposed into controller generation, child-query execution and final extraction. API totals capture successful completions and are lower bounds where failed calls return no usage. Direct and SLM baselines are single-call unless retrieval-augmented; fixed operators have zero marginal model cost but nonzero engineering cost. Engineering amortization is an explicitly hypothetical sensitivity (Appendix A), not measured labor or total cost of ownership. Self-hosted BrowseComp costs use active runtime by route with generation-plus-judge total reported separately. Reported statistics and cost totals are computed from stored logs.

3 Results

3.1 Known-operation tasks

Matched routes lead on controlled tasks. On T-STRUCT, Gemma extracts records and a fixed Python operator performs the aggregation, which is exact by construction, so this row measures extraction success and cost. It reaches 100% accuracy versus 96.7% for RLM at about $99\times$ lower captured deployed-configuration API cost; because both are near ceiling, the result establishes cost dominance only, and supports neither a reliable accuracy difference nor a pure matched-model scaffold-overhead estimate. T-MIXED requires semantic judgments. A matched Gemini ablation separates finalization from semantic error: the count-only finalizer returns no valid answers while the JSON finalizer returns valid answers on all 20 examples and solves 13. A frozen $N=30$ endpoint compares direct Gemini with JSON-finalized RLM: direct Gemini solves 28/30 at \$0.0223, RLM solves 18/30 at \$0.1089 (paired exact $b=12$, $c=2$, $p=0.0129$) and remains about $4.9\times$ costlier. We treat T-NLP only as an interface diagnostic because no official RLM accuracy run is logged; moreover, the classification target requires categorical labels while the current aggregation scaffold assumes numeric-style answers.

Oolong specialization persists across labels and sources. On label-provided LP-80, RLM reaches 82.4% mean score against 70.0% for a generic fixed-parser/SLM fallback, although that difference

Table 3: Main route results and boundaries. LP-80 is mean score; LP-150 and all other accuracy entries are exact. “Best non-RLM” prioritizes accuracy on T-MIXED. T-STRUCT compares deployed configurations; CD-45 is diagnostic. Equal-cap BrowseComp active-runtime cost is \$13.279 (\$15.015 including lower-bound interruption billing and judging).

Role / regime	Best non-RLM	Standard RLM	Boundary supported by this row
T-STRUCT controlled aggregation	Gemma extract + fixed agg.: 100%, \$0.0002/ex	96.7%, \$0.0198/ex	Near-ceiling accuracy; about 99× lower captured cost in this cross-model comparison. This does not isolate scaffold overhead.
T-MIXED controlled count	Direct Gemini: 28/30, \$0.0223 total	RLM: 18/30, \$0.1089 total	Same-model direct inference wins on the frozen $N=30$ endpoint; the $N=20$ finalizer ablation isolates answer-contract failure.
Oolong CD-45 mechanism controls	Correct-family SLM+typed: 36/45; wrong-family: 2/45	Same-tools RLM: 31/45 or 29/45; GPT-5/GPT-5.6 Luna: 23/45, 26/45; at least 2.3× costlier	Choosing the operator family matters here; shared tools and tested controllers do not reverse the ordering.
Oolong LP-80/LP-150 aggregation-only	Typed ops: 100% / 97.3%	RLM: 82.4% / 84.7%	Provided labels isolate aggregation and answer formatting; these are not end-to-end results.
Oolong SU-150/VC-45 end-to-end	SLM-labeler + typed ops: 98/150, \$0.839; unused-source VC-45 35/45, \$0.312	RLM 68/150, \$4.145; VC-45 28/45, \$1.139	SU-150 reuses contexts. VC-45 tests new sources: its exact-rate difference includes zero, while its mean-score difference does not.
BrowseComp equal-cap primary ($N=45$)	Text: 37/45, \$1.400; iterative: 33/45, \$1.710; BM25: 25/45, \$0.296	RLM: 31/45, \$9.873	All three RLM comparisons remain unresolved after Holm correction; RLM costs 5.8–7.1× more than iterative/text routes.

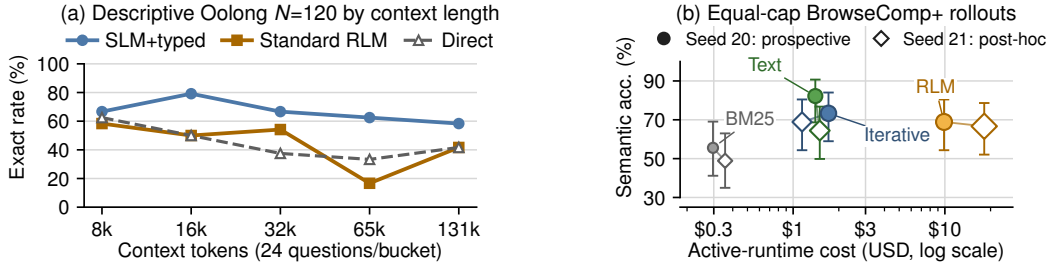


Figure 2: Accuracy and cost across task regimes. (a) Descriptive exact rates on standard-input Oolong development data ($N=120$; 24 per bucket). The inferential contrast is the paired cluster-resampled difference reported in the text, not a per-cell interval. (b) Accuracy versus active-runtime cost on the matched-Qwen equal-cap endpoint (filled circles: prospective seed 20; hollow diamonds: same-query seed 21; $N=45$ each). Whiskers are Wilson intervals, and larger markers mean more all-row model calls (RLM/iterative/text/BM25: 476/426/311/42 and 667/276/312/42). No RLM contrast survives Holm correction in either rollout.

is uncertain ($p=0.089$); its largest gain is on timeline questions and deterministic parsing remains perfect on counting. Gold-label typed operators reach 100% on the same split, so RLM is not the strongest LP-80 route. On frozen held-out LP-150, RLM reaches 84.7% exact while typed rules fixed before evaluation reach 97.3% ($b=23, c=4, p<0.001$). Gold local labels make these aggregation/output-schema ablations, not end-to-end Oolong results. They show that RLM handles flexible schemas better than a generic parser but not that open-ended Python is uniquely required when the schema family is known.

All central standard-unlabeled checks use `context_window_text` without gold local labels. An $N=120$ development replication established the $SLM>RLM$ direction across every 8k–131k bucket (Figure 2(a)), with a paired 22.5-point exact-rate advantage ($b=37, c=10, p=9.8\times 10^{-5}$; 72-cluster interval 11.9–33.0). The fixed pipeline was evaluated on SU-150, CD-45 and pre-output-locked VC-45; before scoring, one parser repair was applied only to rows returning no call, a condition visible without labels.

On SU-150, SLM+typed solves 98/150 at \$0.839, versus RLM 68/150 at \$4.145 and direct Gemini 65/150 at \$1.939. Discordance favors SLM over RLM ($b=44$, $c=14$, $p=1.00 \times 10^{-4}$); no direct–RLM difference is detected ($p=0.788$). The 73-cluster interval gives a 10.6–29.9 point SLM advantage but all rows reuse previously evaluated context windows so SU-150 is scale/method-lock evidence only.

VC-45 removes the known source-corpus overlap: its official validation rows come from spam and trec_coarse, absent from the test split and all previous Oolong evaluations. A replay of the unchanged question parser agrees with the recorded task and answer-type metadata on all 150 SU-150, 45 CD-45 and 45 VC-45 rows. This validates the parser on these benchmark templates, not on arbitrary paraphrases or unseen operator families. SLM+typed solves 35/45 at \$0.312, versus RLM 28/45 at \$1.139 and direct Gemini 23/45 at \$0.658. SLM–RLM discordance is $b=10$, $c=3$, $p=0.092$ (Holm $p=0.185$); the 20-cluster exact-difference interval is 0.0–31.1 points while the mean-score interval is 3.0–30.9. RLM reaches 13/15 on user questions but 6/15 on counting. Task and source are not factorially balanced, so subgroup differences are descriptive. The specialized route remains cheaper and directionally favored under source-corpus transfer but VC-45 alone establishes neither a significant exact-rate advantage nor document-level independence.

CD-45 retains the specialization controls: using the wrong operator family collapses the typed route from 36/45 to 2/45. Giving standard RLM the same operator library yields 31/45 with the original-only tools or 29/45 with atomic tools, but remains at least $2.3 \times$ costlier. An independent standard-RLM repeat drops from 27/45 to 21/45. Together, SU-150 supplies scale, VC-45 supplies source-corpus transfer and CD-45 isolates specialization and same-tools behavior; none is a full Oolong leaderboard.

Tested controllers preserve the Oolong ordering. Strict-stop GPT-5 and GPT-5.6 Luna standard-RLM runs solve 23/45 and 26/45 CD-45 cases, below SLM+typed’s 36/45 (descriptive paired $p=0.00235$ and $p=0.041$). On CD-15, the official trained Qwen3-30B-A3B policy solves 10/15 and unmodified public λ -RLM code with Qwen3-8B solves 8/15, versus 15/15 for SLM+typed. The λ run counts two 1,800-second timeouts incorrect. These sensitivities do not reverse the observed ordering, but differing populations and fidelity tiers preclude pooling. They support a limited robustness claim, not full official-family benchmark parity (Appendix A).

3.2 Adaptive search

Stronger alternatives leave the comparison unresolved. BrowseComp-Plus supports controlled retrieval/agent comparisons. We separate three stages: an exploratory 80-query study, a first untouched 45-query replication and the prospective equal-cap comparison below. The exploratory study gave RLM/text/BM25 54/48/39 correct, but completion across two episodes after partial score visibility limits its evidential role. On the first untouched shard, the same routes scored 33/29/21. RLM exceeded BM25 by 26.7 points (paired 95% interval 11.1–44.4; Holm $p=0.0151$), while its 8.9-point lead over text remained uncertain (−6.7–24.4). RLM completed 42/45, cost \$6.746 and had a 425.7-second p95 latency; text completed 45/45 at \$1.610 and 44.9 seconds. A blinded second-model audit agreed on 128/129 successful outputs ($\kappa=0.983$). The positive RLM–BM25 result is retained; it does not establish superiority to adaptive search.

The prospective equal-cap endpoint tests four routes on a second untouched 45-query shard. We added iterative lexical search and attempted all 180 Qwen3.6 route cells under common 20-decision/1,200-second caps before semantic scoring. RLM scores 31/45 (68.9%), versus iterative 33/45 (73.3%), text 37/45 (82.2%) and BM25 25/45 (55.6%). Differences are −4.4, −13.3 and +13.3 points, with discordances 6/8, 5/11 and 10/4. Paired 95% intervals are [−20.0, 11.1], [−31.1, 4.4] and [−2.2, 28.9] points; no contrast survives Holm correction ($p_{\text{Holm}} \geq 0.539$). The second judge closely agrees ($\kappa=0.985$).

RLM completes 41/45 with 50.1/628.8-second median/p95 latency. Iterative/text complete all 45 with 14.1/90.2 and 22.1/47.6 seconds; BM25 completes 42/45. On 39 common completions, scores are 30/28/31/22; this post-treatment sensitivity leaves the adaptive routes near parity. A question-only Qwen leakage probe scores 0/45 under both judges: neither finds a correct answer without the corpus. This endpoint does not establish the earlier RLM–BM25 advantage; because both queries and the resource contract changed, the difference cannot be attributed to equal caps alone; Figure 2(b) plots both equal-cap rollouts against active-runtime cost. Open search alone is not a sufficient routing signal.

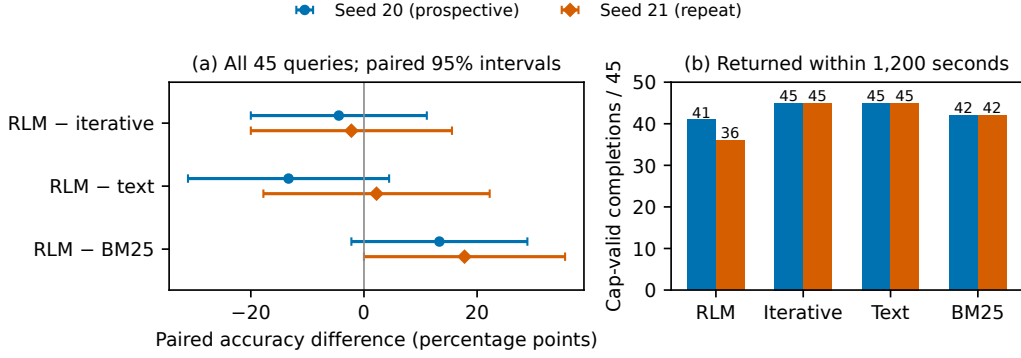


Figure 3: Paired accuracy and completion on the same 45 search queries. (a) RLM minus each alternative, with paired 95% bootstrap intervals conditional on each rollout; these marginal contrast intervals are not simultaneous confidence bands. Holm-adjusted tests resolve no RLM advantage. (b) Outputs returned within the 1,200-second cap. The repeat has 37 successful RLM executions but only 36 cap-valid completions. Dropping incomplete cases changes the comparison population.

Rollouts and completion subsets answer different questions. A post-hoc seed-21 repeat also leaves the RLM comparisons unresolved. RLM/iterative/text/BM25 complete 37/45, 45/45, 45/45 and 42/45 and score 30/31/29/22; no RLM contrast survives Holm ($p_{\text{Holm}} \geq 0.346$). A cap-valid completion finishes within 1,200 seconds: RLM has 36/45, and on 34 rows cap-valid for every route the scores are 28/23/20/18. This post-treatment subset puts RLM first, but selecting cases by completion changes the population; it does not identify a causal effect of attrition. RLM costs 12.1–15.9 $\times$ the adaptive routes. Sol agrees on 166/169 outputs ($\kappa=0.961$); the repeat is not pooled with the prospective endpoint.

On seed 21, RLM trails iterative search by 2.2 points over all 45 queries but leads by 14.7 points on the 34 common cap-valid completions: a 16.9-point change from selection alone. For text, the corresponding differences are +2.2 and +23.5 points. A paired comparison of the RLM–text difference across seeds changes by 15.6 points, with a case-bootstrap interval of $[-8.9, 40.0]$. This descriptive interval is conditional on the two observed rollouts; it does not establish a population-level seed effect. Figure 3 displays the all-case comparisons and their completion denominators.

Implications for route choice. Known operation families warrant fixed-operator and semantic-worker baselines. Open search warrants strong retrieval and adaptive nonrecursive alternatives. Our results support these comparisons, not an automatic router: flexible schemas do not uniquely require generated Python, and open search alone does not establish an RLM advantage.

4 Mechanism Analysis

Failures occur in task interpretation, evidence selection, aggregation, tool use and finalization rather than one RLM component. Appendix C.1 groups these observations by known-operator overhead, answer-contract sensitivity, semantic-local error, output-schema rigidity, search instability, context constraints and multiple-choice bias.

Additional orchestration need not help. A strong LLM can delegate local interpretation to cheaper workers [Narayan et al., 2025], but this adds little when the operation is already fixed. In the T-NLP diagnostic, LLM+SLM scores 83.3% versus 87.5% for SLM direct at about 3.4 $\times$ the cost. This supports checking a direct worker before adding orchestration; it is not an official RLM comparison.

Traces expose failures at distinct stages. CD-45 case 710070026 illustrates a typed-route failure: all seven local labels are correct, yet the route returns “contradiction” rather than the gold least-frequent label “neutral”; RLM and direct inference are correct. Thus accurate local labeling does not guarantee correct aggregation or tie handling. Conversely, case 312030020 is solved by RLM and the typed route while direct inference returns the wrong user. These outcome-selected examples

242 explain distinct failure stages, not their prevalence (Appendix C.2). Stored T-MIXED traces likewise
243 show successful dual-gate filters but also vocabulary overreach and brittle finalization. These traces
244 localize where failures surface; they do not attribute them to the scaffold as a whole.

245 **Multiple-choice position bias limits the QA diagnostic.** LongBench-v2 Academic QA is multiple-
246 choice so answer-position priors confound retrieval-vs.-RLM comparisons. On the $N=50$ discovery
247 set, canonical BM25 predicts A 68% of the time while gold-A is 18%. Rotating priming improves
248 pooled $N=94$ ($p=0.015$) but not held-out $N=44$ ($p=1.0$); single-call accuracy is unmeasured. We
249 therefore keep it as a validity diagnostic, with prior debiasing comparisons in Appendix B.

250 5 Related Work

251 RLM externalizes context through controller-generated programs and recursive calls [Zhang et al.,
252 2026]. Follow-ups test whether recursion drives gains or add typed operators, coding agents and
253 recursive harnesses [Alizadeh et al., 2026, Roy et al., 2026, Ehrlich and Blackman, 2026, Cao et al.,
254 2026, Lumer et al., 2026]; Wang reports task-dependent depth, cost and strong-model degradation
255 [Wang, 2026]. Unlike trajectory studies, our unit is the complete route, compared on matched rows
256 and scorers with paired uncertainty.

257 Cost-aware agent evaluation measures answers and the resources used to obtain them [Kapoor et al.,
258 2025, 2026, Bragg et al., 2026]; RAG inference scaling allocates a given input-token budget [Yue
259 et al., 2025]. We apply both ideas to complete long-context routes under explicit information, scoring
260 and budget contracts and test recommendation stability across rollouts.

261 Oolong separates local labeling from aggregation, which supports our mechanism ablations [Bertsch
262 et al., 2026]. Long-context suites test reasoning, effective context and retrieval/length effects [Bai
263 et al., 2025, Chen et al., 2026c, Hsieh et al., 2024, Yen et al., 2025, Zhang et al., 2024, Modarressi
264 et al., 2025, Li et al., 2025]. Decomposition, search and routing systems select evidence or execution
265 routes [Xu et al., 2026, Song et al., 2026, Ong et al., 2025, Li et al., 2024, Chen et al., 2026a];
266 search-space scale can also change lexical-versus-agentic ordering [Wang et al., 2026]. Our claims
267 remain limited to the declared fixed pools.

268 6 Limitations

269 **External validity.** We diagnose task-conditional failure modes, not average long-context perfor-
270 mance. VC-45 transfers source corpora but not documents because Oolong lacks IDs; SU-150 reuses
271 windows. Official-family evidence covers CD-15 and the controlled curve ends at 131k; longer rows
272 are stress tests.

273 **Measurement.** The 180 equal-cap cells run under fixed caps across three execution episodes. Cross-
274 family judge agreement and a question-only leakage control assess stability, not human validity; there
275 is no human adjudication amid known contextual judge fragility [Tan et al., 2025, Xu et al., 2025].

276 **Cost and scope.** Costs exclude engineering and TCO; alternatives are not exhaustive and recursion
277 depth is fixed. BrowseComp bins as small as $n=10$ preclude subgroup claims. The below-RLM
278 same-row embedding rerank is not full-corpus dense retrieval (Appendix B). Seven repeat RLM calls
279 exceed the cap and score incorrect; self-hosted and API dollars remain separate. Automated routing
280 could suppress useful methods in untested domains.

281 **Ethics and AI assistance.** Public and synthetic benchmarks involve no human subjects. AI tools
282 assisted coding, experiment orchestration and editing; the authors verified all outputs.

283 7 Conclusion

284 Specialized routes lead on tested known operations at lower cost, while equal-cap search yields
285 no RLM advantage despite $5.8\text{--}15.9\times$ higher cost. Completion filtering can change the winner,
286 so recursion should be judged against operation-matched alternatives using all-query accuracy,
287 completion and cost.

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417 A Protocol Details

418 **Cost-aware audit protocol.** Table 4 summarizes the protocol used throughout this study. Because
 419 the audited object is a scaffolded inference procedure, a row should not be promoted from diagnostic
 420 evidence to method evidence unless the input contract, output schema, scorer, failure accounting and
 421 cost trace are matched.

422 **Role-based model coverage.** The primary panel assigns models to deployment roles rather than
 423 constructing a broad leaderboard: Gemini 2.5 Flash controls the logged standard RLM and supplies
 424 direct frontier inference, Gemma 4 26B-A4B supplies SLM worker paths and GPT-4o-mini controls
 425 the short T-NLP LLM+SLM route. GPT-5 is used for the T-STRUCT frontier-controller exception
 426 and a CD-45 controller sensitivity. Many central alternatives are deterministic or structural so the
 427 unresolved breadth question is same-contract coverage of stronger RLM-family systems rather than
 428 additional direct-LLM columns.

429 That stronger-family evidence includes frozen-manifest adapter pressure tests, matched standard-
 430 unlabeled smokes, the CD-15 repeatability study and CD-45 same-operator-library diagnostics.
 431 The unmodified public λ -RLM repository attempts all 15 frozen CD-15 IDs with its published
 432 Qwen3-8B settings and reaches 8/15 exact with two declared timeouts. MIT OASYS’s official
 433 `r1m-qwen3-30b-a3b-v0.1` post-trained policy reaches 10/15 exact with one timeout through the
 434 canonical `r1m` harness and model-card Oolong settings. The λ row is official-code cross-task evidence
 435 rather than an original-paper reproduction; rankable official SRLM and same-contract RAH rows
 436 remain unavailable.